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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	疯狂星期四
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4, SDG 8, SDG 9, SDG 10
Project / Solution Name	CareerNovacoke
Name (Leader)	LIN JIAXI
Email & Phone (Leader)	breathfreea@outlook.com , 1123587027
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing the job market quickly, but many people cannot keep up with these changes. Workers may lose their jobs if their skills are no longer needed, while fresh graduates and career changers may not know which careers to choose or what skills to learn. Employers also struggle to know which skills their workers need for the future.

This problem affects workers, job seekers, students, businesses, and educational institutions. Without proper guidance and training, people may face unemployment and lower incomes, while businesses may struggle to find skilled workers. At the same time, a lack of skilled talent may slow the growth of important industries such as clean energy, healthcare, and smart agriculture.

We need an AI-powered platform that tracks job market changes, helps people find suitable future careers, and provides personalised learning paths. By connecting future skills and careers to the United Nations' 17 Sustainable Development Goals (SDGs), we can help people prepare for the future while supporting a fairer and more sustainable world.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

CareerNovacoke is an AI-assisted career guidance platform designed to help students, fresh graduates, job seekers, and career changers find suitable career paths in an AI-driven world.

As technology changes the job market, many people are unsure which careers to choose and what skills they need to remain employable. CareerNova aims to make career planning easier by providing personalised recommendations based on each user's interests, skills, salary expectations, and preferred work location.

Users complete a short questionnaire, and the system compares their answers with a curated database of

career information to recommend suitable roles. The platform also provides career details, salary ranges, target countries, certification requirements, learning resources, and personalised learning plans to help users identify skill gaps and take practical steps towards their career goals.

Our current prototype focuses on questionnaire-based career matching and structured career information. In future development, we aim to improve the system with AI tools to analyse changing job market trends and provide more up-to-date career and skill recommendations.

Through the SDC Phase 1 workshops, we aim to apply relevant technical skills such as AI-assisted development, data organisation, database design, and cloud deployment. We use Replit to develop and host our web application, while GitHub supports code management and team collaboration.

Our goal is to make career guidance more accessible and practical, helping people make informed career decisions, develop relevant skills, and prepare for future employment.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: HTML, CSS and JavaScript are used to build the user interface, with Jinja2 templates rendering dynamic web pages.

Backend: Python 3.13+ and Flask 3.1.3+ handle survey submissions, input validation, session management and page navigation.

Career Matching Logic: A Python-based rule-scoring system compares users' skills, interests, salary expectations and preferred countries against 10 predefined career profiles. It ranks career recommendations and generates personalised learning plans using predefined learning resources, certifications and practical activities.

Data Management: Career profiles, survey questions, skills, interests and learning resources are stored in Python data structures. Flask sessions maintain users' survey answers during the questionnaire process.

Development and Deployment: Replit is used to run and host the web application, while GitHub is used for source-code management.

APIs: The current prototype does not use external APIs.

Official SDC Sponsor-Provided GenAI API or Cloud Platform

No official SDC sponsor-provided GenAI API or Cloud Platform is integrated into the current prototype. Replit is used for application hosting.Replit hosts the application, while GitHub manages the source code.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied design-thinking principles by focusing on the challenges faced by fresh graduates, job seekers, and career changers who struggle to identify suitable careers and understand which skills they need to improve.

Empathise and Define: We identified the need for accessible career guidance that connects users' existing skills and interests with suitable career options and practical learning steps.

Ideate: We designed a short, structured questionnaire to collect information about users' skills, interests, salary expectations, preferred countries, and career goals.

Prototype: We developed CareerNova as a web application using Python and Flask. The current prototype includes questionnaire navigation, rule-based career matching, ranked recommendations, and personalised learning plans based on predefined career information and learning resources.

Test and Improve: We use a modular structure to make the application easier to maintain and improve. We will test the questionnaire flow, input validation, matching results, and learning-plan generation, then refine the interface based on user feedback.

Deployment Strategy: We use Replit to run and host the application and GitHub for version control. Before Phase 2, we aim to verify the main user journey, fix identified issues, and ensure the deployed prototype works reliably during the demonstration.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.4: By 2030, substantially increase the number of youth and adults who have relevant skills for employment, decent jobs and entrepreneurship. SDG 8.6: Reduce the proportion of young people not in employment, education or training (NEET). SDG9.5 – Enhance scientific research and upgrade technological capabilities: CareerNova uses a digital platform and a career-matching algorithm to provide career recommendations and personalised learning plans through technology. SDG10.2 – Promote universal social, economic and political inclusion: CareerNova provides structured career information and learning recommendations to help users from different educational and career backgrounds explore suitable career opportunities.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	According to the United Nations Sustainable Development Goals Report 2025, 20.4% of young people aged 15–24 worldwide were not in employment, education or training in 2024. This highlights the importance of accessible career guidance and skills development opportunities. This is a global contextual baseline, not a measured rate among CareerNova users.
Core Measurable KPI	Career guidance task completion rate (%): The percentage

Metric Category	Your Response
The specific unit you will track *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	of usability-test participants who independently complete the main user journey, from answering the questionnaire to viewing ranked career recommendations and a personalised learning plan.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 80% task completion among 15 usability-test participants by the final prototype presentation (at least 12 out of 15 participants). The target will be measured through structured user testing.

RESOURCES:

- United Nations, The Sustainable Development Goals Report 2025 — Goal 8. Official UN report (PDF)
- United Nations, Goal 4: Quality Education. Official SDG 4 targets
- United Nations, Goal 8: Decent Work and Economic Growth. Official SDG 8 targets

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [×] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [√] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [√] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [√] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Flask 3.1.3	Python web framework	BSD-3-Clause
Jinja2 3.1.6	HTML template engine	BSD-3-Clause
Werkzeug 3.1.9	WSGI utilities and web request handling	BSD-3-Clause
Click 8.5.0	Command-line utilities	BSD-3-Clause
ItsDangerous 2.2.0	Signing and security utilities	BSD-3-Clause

MarkupSafe 3.0.4	HTML escaping and safe string handling	BSD-3-Clause
Blinker 1.9.0	Signal support	MIT
Playwright 1.64.1	Browser automation and testing; development dependency	Apache-2.0
greenlet 3.5.6	Dependency of Playwright	MIT OR PSF-2.0
pyee 14.0.0	Dependency of Playwright	MIT
typing-extensions 4.16.0	Python typing compatibility support; dependency of pyee	PSF-2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

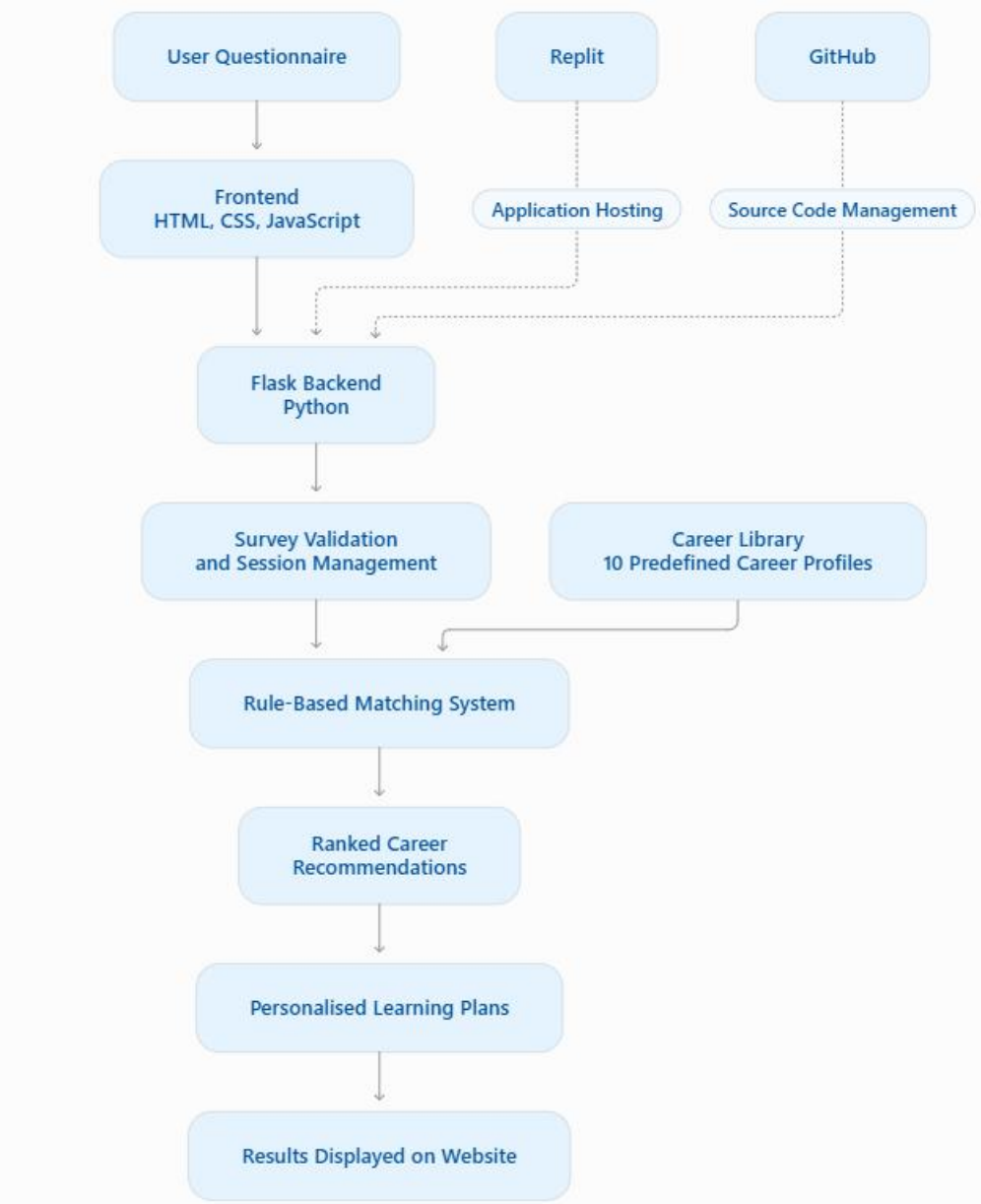
☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

*Appendix

System Architecture

Users complete a questionnaire through the frontend. The Flask backend processes and validates their answers, then passes them to the Python matching system. The matching system compares the answers with the predefined career library, calculates matching scores, ranks career recommendations and generates personalised learning plans. The results are then displayed on the website.



7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

24.10.2026